AQUILA: An Intelligent Geospatial Decision-Support Platform for Urban Flood Risk Assessment and Emergency Dispatch in Manikonda, Hyderabad

V. Saicharan Reddy, A. Sai Santhosh, and B. Shashank

Abstract Rapid, unplanned peri-urbanization in Indian metropolitan regions has severely disrupted natural drainage networks, amplifying the frequency and devastating impact of urban pluvial flash flooding [1], [3]. Conventional disaster management mechanisms often rely on fragmented municipal datasets, static drainage surveys, and uncoordinated emergency response protocols that lack real-time spatial intelligence [2], [14]. This paper presents **AQUILA**, an end-to-end geospatial decision-support prototype designed for urban flood risk intelligence, infrastructure vulnerability monitoring, and optimized emergency dispatch. Centered on the flood-vulnerable Manikonda catchment in the Hyderabad Metropolitan Region (Telangana, India), the platform unites multi-source open geospatial data: a 30m Copernicus Digital Elevation Model (DEM) [5], 10m ESA WorldCover land use rasters [6], OpenStreetMap vector networks [7] (comprising 6,659 road links, 21 drainage canals, and 66 critical infrastructure facilities), and localized meteorological precipitation records [8]. At the core of AQUILA is an Analytical Hierarchy Process (AHP) risk engine whose pairwise comparisons demonstrate strict mathematical consistency ($\text{Consistency Ratio } \text{CR} = 0.0000 < 0.10$). The engine evaluates five normalized hazard drivers—Precipitation Intensity (40%), Terrain Vulnerability (20%), Drainage Proximity (20%), Land Cover Imperviousness (10%), and Historical Flood Recurrence (10%)—with dynamic weight renormalization for missing data layers. For emergency logistics, the platform couples Great-Circle Haversine proximity evaluation with a hazard-aware Dijkstra routing algorithm over a topologically verified road graph of 23,802 vertices and 48,391 edges (94.7% connected component integrity) [10], [11]. The model is evaluated through operational storm simulation (50 mm/hr convective rain yielding $R = 78.10$) and a *leakage-sensitivity ablation experiment* excluding historical recurrence data ($w_H = 0$). The physical ablation model demonstrates that physical terrain, drainage proximity, and soil imperviousness intrinsically drive high-risk identification ($R_{\text{blind}} \ge 75.32$) across five documented flood sites from the October 2020 deluge [2]. Deployed as a high-performance FastAPI service paired with an interactive Folium/Streamlit dashboard, AQUILA establishes an actionable paradigm for municipal disaster management.

Keywords Urban flood intelligence · Geospatial decision support · Multi-factor risk engine · Analytical Hierarchy Process · Manikonda catchment · Hyderabad · Road network graph topology · Emergency dispatch routing

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Case Study: Manikonda Catchment, Hyderabad Metropolitan Region (HMR), Telangana, India.

1 Introduction

Urban pluvial flooding—inundation triggered when extreme localized precipitation overwhelms natural soil percolation and built drainage infrastructure—has emerged as a recurrent crisis across Indian megacities [1], [4]. In rapidly growing metropolitan centers like Hyderabad, Bengaluru, and Chennai, accelerated real-estate development has replaced permeable agricultural soils and natural wetlands with impervious asphalt and concrete surfaces [3], [13]. Furthermore, the historical chain-link lake cascade system (*cheruvus*) and interconnected natural feeder channels (*nalas*) engineered during the Qutb Shahi and Asaf Jahi dynasties have suffered extensive encroachment, fragmentation, and siltation [2], [14].

During the catastrophic October 2020 Hyderabad deluge, several meteorological stations registered in excess of 300 mm of concentrated precipitation within 24 hours, inundating major urban catchments, causing extensive loss of life, and stranding tens of thousands of residents [2]. Post-disaster audits highlighted critical institutional vulnerabilities: response bodies such as the Greater Hyderabad Municipal Corporation (GHMC), the Telangana State Disaster Management Authority (TGSDMA), and the recently established Hyderabad Disaster Response and Asset Protection Agency (HYDRAA) were impeded by fragmented spatial data, lack of real-time road navigability intelligence, and reliance on retrospective damage reports rather than predictive risk modeling [14].

To bridge this operational void, this study introduces **AQUILA**: an academic decision-support prototype engineered to deliver rapid pluvial flood risk evaluation, infrastructure vulnerability assessment, and hazard-aware emergency resource dispatch. The primary contributions of this paper are: (i) an automated, reproducible open-source geospatial acquisition and validation pipeline for peri-urban terrain [5], [6], [7]; (ii) a mathematically consistent AHP multi-factor risk engine with dynamic weight renormalization; (iii) empirical validation contrasting synthetic storm simulations against a leakage-sensitivity physical ablation over historical inundation points [2]; and (iv) a hazard-aware Dijkstra routing model that dynamically penalizes waterlogged street links across a verified 23,802-node road graph [10], [11].

2 Study Area and Geospatial Data Architecture

2.1 Case Study: The Manikonda Pilot Catchment

The focal study domain for AQUILA is the Manikonda catchment, situated in south-western Hyderabad adjacent to the HITEC City and Gachibowli financial corridors ($17.380^\\circ\text{N} - 17.410^\\circ\text{N}$ latitude, $78.360^\\circ\text{E} - 78.400^\\circ\text{E}$ longitude). Manikonda presents an ideal peri-urban case study: high-density multi-story residential enclaves (e.g., Alkapur Township) have proliferated across a rugged granite terrain characterized by an elevation relief of 146 metres (492 m to 638 m above MSL). Steep granite knolls accelerate surface runoff into low-lying retention depressions where historical drainage nalas have been constricted, converting moderate convective storms into acute localized flood events [2], [3].

2.2 Geospatial Data Audit and Raster Dimension Clarification

A critical prerequisite for rigorous hydrological analysis is bounding box integrity. While the primary urban settlement of Manikonda occupies approximately $3.3\text{ km} \times 4.2\text{ km}$, hydrological watersheds are governed by topographical divides rather than municipal borders [3], [12]. Consequently, AQUILA's acquisition pipeline extracts an expanded hydrological buffer envelope ($78.3455^\\circ\text{E} - 78.4194^\\circ\text{E}$ longitude, $17.3606^\\circ\text{N} - 17.4245^\\circ\text{N}$ latitude, spanning approximately $7.85\text{ km} \times 7.07\text{ km}$) to capture upstream ridgeline precipitation draining into the central depression.

At the 1 arc-second ($\sim 30.9\text{ m}$) resolution of the Copernicus GLO-30 DEM [5], this $7.85\text{ km} \times 7.07\text{ km}$ domain translates exactly to **266 columns by 230 rows ($60,940\text{ pixels}$)**, fully resolving the geometric consistency between bounding coordinates and raster grid dimensions. This elevation raster is coupled with

the 10m ESA WorldCover 2020 product [6], OpenStreetMap vector layers [7] (6,659 road segments, 21 drainage canals, 24 lakes/tanks, 66 infrastructure assets), and IMD historical storm records [8].

3 System Architecture and Mathematical Formulation

3.1 Multi-Factor Rule-Based Risk Engine and AHP Weight Derivation

Rather than employing 'black-box' deep neural networks that require massive historical flood-depth training sets rarely available in Indian peri-urban jurisdictions [13], AQUILA establishes an analytically structured, multi-factor rule-based risk model. The composite dimensionless Geographic Risk Index $R \in [0, 100]$ is computed as:

$$R = \frac{\sum_{i=1}^{n} w_i \cdot S_i}{\sum_{i=1}^{n} w_i} = 0.40\, S_P + 0.20\, S_T + 0.20\, S_D + 0.10\, S_L + 0.10\, S_H \quad (1)$$

To guarantee mathematical transitivity across factor weights, Saaty's Analytical Hierarchy Process (AHP) was performed [3], [13]. Constructing the 5×5 pairwise comparison matrix $\mathbf{A}$ structured upon meteorological and hydrological dominance hierarchies in southern India [2], [4]:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 2 & 4 & 4 \\ 1/2 & 1 & 1 & 2 & 2 \\ 1/2 & 1 & 1 & 2 & 2 \\ 1/4 & 1/2 & 1/2 & 1 & 1 \\ 1/4 & 1/2 & 1/2 & 1 & 1 \end{bmatrix} \quad \longrightarrow \quad \mathbf{w} = \begin{bmatrix} 0.40 \\ 0.20 \\ 0.20 \\ 0.10 \\ 0.10 \end{bmatrix} \quad (2)$$

Evaluating the principal eigenvalue yields $\lambda_{\max} = 5.0000$. The Consistency Index is $\text{CI} = (\lambda_{\max} - n) / (n - 1) = 0.0000$. Using the standard Random Consistency Index $\text{RI} = 1.12$ for $n = 5$, the Consistency Ratio is $\text{CR} = \text{CI} / \text{RI} = \mathbf{0.0000} < 0.10$. While this mathematical result does not claim that the initial pairwise criteria are free from expert subjectivity, it verifies that the resulting weights preserve strict mathematical transitivity and internal consistency across the decision matrix.

• **Precipitation Hazard (S_P, $w_P = 0.40$):** Assigned highest weight as the primary dynamic trigger. Rainfall intensity is normalized piece-wise using IMD thresholds [8]: light ($<15.6\text{ mm/hr}$), heavy ($15.6 - 64.5\text{ mm/hr}$), very heavy ($64.5 - 100\text{ mm/hr}$), and cloudburst ($\ge 100\text{ mm/hr} \rightarrow S_P = 100.0$).
• **Terrain Vulnerability (S_T, $w_T = 0.20$):** Evaluated as $S_T = 0.60\, S_{\text{elev}} + 0.40\, S_{\text{slope}}$, where slopes $<1.5^\circ$ receive $S_{\text{slope}} = 100.0$ (severe water stagnation risk) and elevation is normalized across the 492–638m relief.
• **Drainage Proximity (S_D, $w_D = 0.20$):** Models backwater surcharge and overtopping from open nalas [12]. Distance decay: $d \le 50\text{m} \rightarrow 95.0$; $50 < d \le 150\text{m} \rightarrow 75.0$; $150 < d \le 300\text{m} \rightarrow 45.0$; decaying to 10.0 beyond 600m.
• **Land Cover Imperviousness (S_L, $w_L = 0.10$):** Calibrated from ESA WorldCover classes [6]: Class 50 (Built-up urban: $S_L = 90.0$), Class 80 (Water buffer: 95.0), Class 10 (Tree cover: 15.0).
• **Historical Flood Recurrence (S_H, $w_H = 0.10$):** Incorporates spatial memory of verified chronic inundation locations: $d \le 150\text{m} \rightarrow 100.0$; $150 < d \le 300\text{m} \rightarrow 75.0$.

3.2 Dynamic Weight Renormalization under Missing Data

A critical vulnerability in static multi-criteria engines is score collapse when a data layer is unavailable. AQUILA dynamically detects missing inputs and renormalizes active weights: $R = (\sum_{i \in \text{Active}} w_i S_i) / (\sum_{i \in \text{Active}} w_i)$. For example, if satellite land cover is corrupted ($w_L = 0.10$ missing), the active sum is 0.90, renormalizing rainfall to $w_P^* = 0.40 / 0.90 = 0.444$. This ensures that risk evaluation remains robust and operational during sensor outages.

3.3 Topological Road Network Graph Modeling

The road network is modeled as a directed topological graph $G = (V, E)$ using NetworkX [10], [15]. Intersections and terminal points define vertices (V), quantized to 6 decimal places ($~0.1\text{m}$ precision). Segments define edges (E) weighted by projected ellipsoidal length. Graph analysis confirms **23,802 vertices** and **48,391 directed edges**. Weakly connected component analysis establishes that the largest vehicular connected component contains **22,531 nodes (94.7% of the total network)**, comfortably exceeding the engineering reliability target of $\ge 90\%$.

3.4 Hazard-Aware Emergency Routing and Road Closure Avoidance

Unlike standard navigation systems that route solely on static distance or dry-weather travel time, AQUILA implements a hazard-aware dispatch engine. When an emergency incident is reported, resource allocation proceeds via a two-tier strategy:

1. **Nearest-Depot Discovery:** Evaluates Great-Circle Haversine distance across municipal depots (`aquila_mvp.db`), filtering assets by capability mapping (e.g., `WATERLOGGING` $\rightarrow$ high-volume dewatering pumps; `STRUCTURAL_HAZARD` $\rightarrow$ rescue skiffs). Candidate assets are subsequently verified via full road-network traversal to guarantee that the selected depot is also the fastest navigable route.

2. **Dynamic Flood-Impeded Dijkstra Routing:** Edge traversal costs on graph G are dynamically modulated by midpoint risk: $\text{Cost}(e) = \text{Length}(e) \times [1 + \alpha \cdot (R_{\text{segment}} / 100)]$, where $\alpha = 5.0$. Furthermore, road segments exhibiting $R_{\text{segment}} \ge 70.0$ (`HIGH` risk / severe nala overtopping) are dynamically pruned from the active graph. If graph pruning temporarily disconnects a destination enclave, the solver activates a fallback heuristic identifying the highest elevated peripheral road access point and alerts dispatch operators that amphibious water-rescue assets are required.

Table 1 Geospatial data inventory and processing specifications for the Manikonda study catchment

Data Layer	Primary Source	Resolution / Features	Processing Workflow
Digital Elevation Model	Copernicus GLO-30 [5]	30m grid (230×266 px)	7.85km × 7.07km watershed buffer clip, NumPy central difference slope
Road Network Graph	OpenStreetMap via Overpass [7]	6,659 segments; 23,802 nodes	Coordinate quantization (6 dec), topological directed graph [10]
Drainage & Waterways	OSM Waterways [7]	21 canals, 24 lakes/ponds	Polyline extraction, Euclidean distance buffering (50–600m)
Land Cover Permeability	ESA WorldCover 10m [6]	10m raster (888×768 px)	Class mapping to runoff coefficients (Classes 10 to 80)
Critical Facilities	OSM Infrastructure [7]	66 facilities, 5 stations	Facility tagging, nearest emergency resource indexing
Historical Inundation	GHMC DRF / TSDPS Reports [2]	5 validated ground points	Field disaster survey coordinates from Oct 13–15, 2020 deluge

Table 2 Multi-factor risk engine parameter weights, scoring rubrics, and physical significance

Risk Component	Weight (w_i)	Input Metric	Normalization Rubric ($S_i \in [0, 100]$)	Physical Mechanism
Precipitation (S_P)	0.40	Rainfall rate (mm/hr)	Linear piece-wise (IMD thresholds: 15.6, 64.5, 100 mm/hr [8])	Primary pluvial storm inundation volume
Terrain (S_T)	0.20	Elevation (m) & Slope ($^\circ$)	$0.60 S_{\text{elev}} + 0.40 S_{\text{slope}}$ ($<1.5^\circ \rightarrow 100$, $>15^\circ \rightarrow 15$ [9])	Gravity runoff accumulation in valley sinks
Drainage Proximity (S_D)	0.20	Distance to nala (m)	Stepwise: $\le 50\text{m} \rightarrow 95$; $50–150\text{m} \rightarrow 75$; $>600\text{m} \rightarrow 10$	Channel overtopping and backwater surcharge [12]
Land Cover (S_L)	0.10	ESA WorldCover class	Class 50 (Built-up: 90); Class 80 (Water: 95); Class 10 (Tree: 15)	Impervious runoff generation vs. infiltration [3]
Historical Flooding (S_H)	0.10	Distance to flood point (m)	Stepwise: $\le 150\text{m} \rightarrow 100$; $150–300\text{m} \rightarrow 75$; $>1000\text{m} \rightarrow 5$	Empirical recurrence at chronic flood hotspots [2]

Table 3 Empirical performance validation benchmarks across AQUILA platform modules

System Module	Target Specification	Observed Benchmark	Verification Status
Road Network Connectivity	Largest Connected Component $\ge 90\%$	**94.7%** (22,531 / 23,802 nodes connected)	PASS [10]
Geospatial Validation Audit	13 automated data integrity checks	**13 / 13 checks passed** (100% compliance)	PASS
FastAPI Hazard Evaluation Latency	Composite risk score compute $<500\text{ ms}$	**18.4 ms** per query coordinate	PASS
Haversine Resource Dispatch	Nearest-depot search time $<100\text{ ms}$	**2.1 ms** across active resource inventory	PASS
Streamlit Dashboard Responsiveness	Interactive map render $<2000\text{ ms}$	**840 ms** cold start, sub-200ms reactive update	PASS

Table 4 Documented October 2020 inundation sites, ground provenance, sub-scores, and blind physical risk evaluation ($w_H = 0$)

Incident ID & Location	Coordinates	Source Authority Provenance	Depth (m)	Sub-Scores (S_P, S_T, S_D, S_L)	Blind R_{blind}
FLD-01: Alkapur Township Sec 1	17.3912°N, 78.3845°E	GHMC Monsoon Squad Field Log [2]	0.55m	64.6, 91.3, 75.0, 90.0	**75.67** (HIGH)
FLD-02: Puppalguda Main Road	17.4015°N, 78.3750°E	Cyberabad Traffic Police Advisory [2]	0.45m	64.6, 87.4, 95.0, 90.0	**79.26** (HIGH)
FLD-03: Shaikpet Trunk Nala	17.4050°N, 78.4020°E	NRSC Bhuvan Flood Layer [2], [14]	0.65m	64.6, 94.9, 95.0, 90.0	**80.91** (HIGH)
FLD-04: Golden Temple Colony	17.3875°N, 78.3780°E	RWA Memorandum & Media Logs	0.50m	64.6, 89.7, 75.0, 90.0	**75.32** (HIGH)
FLD-05: ORR Service Underpass	17.3820°N, 78.3650°E	Telangana Fire Services Dispatch [14]	0.70m	64.6, 96.4, 95.0, 90.0	**81.25** (HIGH)

4 Experimental Evaluation: Synthetic Simulation and Leakage-Sensitivity Analysis

To rigorously evaluate AQUILA, we distinguish between controlled operational storm simulations and empirical leakage-sensitivity experiments [2], [13]:

A. Synthetic Cloudburst Simulation: A high-intensity convective thunderstorm event (50.0 mm/hr continuous rainfall) was simulated over the Alkapur Township Sector 1 depression ($17.3912^\circ\text{N},

78.3845^\circ\text{E}$, elevation 512m, slope 1.2°). The model computed sub-scores: $S_P = 64.62$, $S_T = 91.26$, $S_D = 75.0$ (65m to open nala), $S_L = 90.0$ (built-up impervious), and $S_H = 100.0$ (120m to historical flood point), resulting in a composite risk score of **$R = 78.10$ (`HIGH` risk alert)**. The dispatch module automatically identified the nearest mobile dewatering pump at Manikonda Municipality Depot (2.8 km via elevated road corridors, avoiding the submerged Puppalguda culvert).

B. Leakage-Sensitivity Analysis via Blind Physical Ablation: To evaluate whether the model's high-risk classifications depend circularly upon historical flood proximity memory (S_H), a blind physical ablation experiment was executed by completely setting the historical recurrence weight to zero ($w_H = 0$). Active factor weights were dynamically renormalized ($w_P^* = 0.444, w_T^* = 0.222, w_D^* = 0.222, w_L^* = 0.111$). As detailed in Table 4, evaluating this purely physical model across the five documented October 2020 inundation locations yielded risk scores ranging from $R_{\text{blind}} = 75.32$ to 81.25, with all five locations exceeding the critical threshold ($R_{\text{blind}} \ge 70.0$). While not an out-of-sample validation on independent future catchments, this experiment confirms that high-risk designations are driven intrinsically by hydro-topographical physics (low elevation, flat slopes, and nala proximity) rather than solely by historical proximity memory.

5 Discussion and Practical Limitations

While AQUILA establishes an effective open-source decision-support paradigm, several engineering trade-offs warrant consideration: **(1) DEM Resolution:** The 30-meter resolution of the Copernicus GLO-30 DEM averages micro-topographical barriers (curbs, roadside gutters, compound walls) narrower than 30m; integrating $<1\text{m}$ drone LiDAR would refine street-level flow paths. **(2) Gauge Network Density:** Precipitation relies on four peripheral perimeter stations; localized cloudbursts directly over Manikonda require dense street-level IoT rain gauges. **(3) Non-Authoritative Status:** AQUILA is an academic prototype designed to complement statutory directives from GHMC and HYDRAA [14].

6 Conclusion and Future Directions

This research demonstrated AQUILA, an integrated geospatial platform that translates heterogeneous satellite, vector, and meteorological data into actionable urban flood risk intelligence. By coupling an AHP-calibrated risk engine ($\text{CR} = 0.0000$; confirmed high sensitivity in blind physical ablation across documented October 2020 disaster points [2]) with high-integrity road network graph analysis (94.7% connectivity over 23,802 nodes [10]), the platform bridges the gap between complex environmental data and practical municipal emergency dispatch. Future work will integrate ultrasonic water-level IoT telemetry and 2D shallow-water hydrodynamic flow modeling across the broader Hyderabad region.

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